

Purification of nucleic acids

Risk assessment

- Guanidine salts form TOXIC GASES with bleach or strong acids
- ▷ Wear gloves, safety glasses, lab coat
- DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Jun 10, 2023

This is a bench card. Full protocol available online.

A > Purification by alcohol precipitation

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| ☐ R0045 3 M Sodium acetate | ☐ Buffer PE (Wash buffer) |
| ☐ Ethanol, ice-cold | ☐ R0056 5 mM Tris hydrochloride |

- (1.) Mix 1.0 vol sample with 0.1 vol 3 M sodium acetate. Mix well.
- (2.) *Optional:* Add 10–20 µg/mL linear polyacrylamide to precipitate picogram quantities of nucleic acids.
- (3.) Add 2.0 vol ethanol to precipitate DNA, or 2.5–3.0 vol ethanol to precipitate RNA. Mix by vortexing.
- (4.) Precipitate for 15–30 min on ice.
- (5.) Centrifuge at 12 000 × *g* for 15 min at 4 °C. Discard supernatant and spin briefly to remove residual solvent. The pellet will appear nearly transparent.
- (6.) Wash pellet with 650 µL Buffer PE or freshly prepared 70% ethanol. Discard supernatant carefully.
- (7.) Dry pellet for 10–15 min at room temperature, or for 1 min in a vacuum centrifuge.

Critical: Avoid overdrying, which impairs dissolution and may denature small dsDNA.

- (8.) Resuspend in 15–30 µL 5 mM Tris pH 8.0 or buffer of choice.

B > Purification over a silica matrix

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| ☐ Buffer PB (Binding buffer) | ☐ R0056 5 mM Tris hydrochloride |
| ☐ Buffer PE (Wash buffer) | |

- (1.) Mix 1.0 vol sample with 2.0 vol Buffer PB.
- (2.) Apply up to 700 µL to a silica spin column. Centrifuge at 11 000 × *g* for 30 s or use vacuum manifold.
- (3.) Wash with 650 µL Buffer PE.

Critical: Let stand for 2–5 min before spinning if DNA will be used in salt-sensitive applications such as blunt-end ligation.

- (4.) Repeat wash to minimize salt carryover.
- (5.) Dry membrane by centrifugation at 11 000 × *g* for 1 min.

Critical: Residual ethanol may persist if flow-through is not discarded prior to drying.

- (6.) Place spin column in a clean tube. Apply 15–30 µL 5 mM Tris pH 8.0 onto the silica membrane. Incubate 1 min. Elute at 11 000 × *g* for 1 min.
- (7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

Recipe (available online) Troubleshooting (available online) Notes (available online)

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